INT395:SUPERVISED LEARNING

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: explain data preprocessing, feature engineering and feature selection techniques for preparing high-quality datasets used in supervised machine learning.

CO2 :: apply supervised classification algorithms and performance evaluation metrics to develop predictive machine learning models.

CO3 :: analyze ensemble learning, hyperparameter optimization and model interpretability techniques to improve prediction performance and transparency.

CO4 :: evaluate regression models using appropriate performance metrics for predictive analytics across diverse application domains.

CO5 :: evaluate classical time series forecasting models based on temporal data characteristics and forecasting accuracy.

CO6 :: apply machine learning pipelines, model validation, deployment and explainability techniques to develop end-to-end supervised learning solutions.

Unit I

Introduction : overview of supervised learning and use cases, types of supervised learning, setting up python and scikit-learn environment, types of data, loading datasets, data exploration with Pandas and matplotlib/seaborn

Data Preprocessing : common data issues, handling missing values and outliers, handling class imbalance, scaling features, encoding categorical variables, Feature engineering, Feature selection, data splitting, dimensionality reduction

Unit II

Classification with scikit-learn : Overview of classification, evaluation metrics- Accuracy, Precision, Recall, F1-Score, ROC-AUC, confusion matrix, perceptron, logistic regression, k-Nearest Neighbors, Decision Tree, Support Vector Machine, Naïve Bayes

Unit III

Ensemble Methods and Hyperparameter Tuning : Overview of ensemble models, how ensembles can reduce error, Bagging, Boosting, Stacking, need for hyperparameter tuning, grid search, random search, Bayesian Optimization

Explainable AI : Need for Explainable AI, Global vs Local Interpretability, Feature Importance, LIME (Local Interpretable Model-Agnostic Explanations), SHAP (SHapley Additive Explanations), Model Transparency and Trustworthiness

Unit IV

Regression with scikit-Learn : Introduction to regression, exploratory data analysis, evaluation metrics, linear regression, RANSAC model, polynomial regression, regularized regression, support vector regression, decision tree regression, random forest regression

Unit V

Time Series Regression : Time series data characteristics, univariate vs multivariate, regular vs irregular intervals, preparing Time Series data, Data splitting strategies, autoregressive models, moving average models, ARIMA, SARIMA

Unit VI

Pipelines and Model Evaluation : Cross Validation strategies, debugging algorithms with learning curves and validation curves, Streamlining workflows with Scikit Learn Pipelines

Model Deployment : importance of model deployment, model serialization and deserialization, local deployment, Web service deployment, Serverless options, GitHub-based Deployment

List of Practicals / Experiments:

List of practical

- ·Decode hidden patterns in passenger survival data by performing comprehensive exploratory data analysis on the Titanic dataset to uncover data quality issues, feature relationships, and meaningful insights for predictive modeling.

- ·Transform raw data into a machine-learning-ready asset by designing a robust preprocessing workflow that handles missing values, categorical variables, feature scaling, class imbalance, and data quality challenges.
- ·Enhance predictive intelligence for real-estate valuation by engineering informative features, selecting relevant attributes, and reducing dimensionality for accurate house price prediction.
- ·Establish a reliable employee attrition prediction framework by validating supervised learning models using appropriate data partitioning and cross-validation strategies.
- ·Recommend fair and data-driven loan approval decisions by evaluating multiple classification models and selecting the most effective predictive solution for financial risk assessment.
- ·Support intelligent medical decision-making by developing and rigorously evaluating a disease diagnosis prediction model using multiple classification performance measures.
- ·Build a transparent fraud detection system by combining ensemble learning with Explainable AI techniques (SHAP and LIME) to justify prediction outcomes and enhance stakeholder trust.
- ·Identify customers at risk of leaving a business by benchmarking ensemble learning techniques and interpreting their effectiveness for customer churn prediction.
- ·Improve predictive excellence through model optimization by systematically exploring hyperparameter search strategies and analyzing their impact on model performance.
- ·Model complex relationships in continuous data by investigating and comparing regression algorithms for accurate predictive analytics.
- ·Forecast future trends from historical observations by constructing and evaluating ARIMA and SARIMA models for real-world time series prediction.
- ·Automate the supervised learning lifecycle by integrating preprocessing, feature engineering, model training, validation, and evaluation into a reusable Scikit-learn pipeline.
- ·Operationalize a machine learning solution by deploying a trained predictive model as an interactive Streamlit application using industry-standard serialization techniques.
- ·Engineer an end-to-end AI-powered decision support system for a real-world application by performing problem formulation, data preparation, predictive modeling, explainability, evaluation, deployment, and presentation of actionable insights.

Text Books:

1. MACHINE LEARNING-I by CHANDRA S.S, VINOD, PHI Learning

References:

1. MACHINE LEARNING by ETHEM ALPAYDIN, MIT Press